

Spoken Tutorial on Understanding APIs

Module	Building with AI
Episode	2. Understanding APIs
Learning Objective	At the end of this tutorial learner will be able to 1. Define what an API is in the context of AI. 2. Explain the role of API keys in secure API access. 3. Describe how applications send requests and receive AI-generated responses. 4. Apply the concept of prompts as inputs for AI models.
Approx. Duration	3-4 min
Outline	• What an API is and why it matters • Prompts as inputs sent through APIs • API keys for authentication and access control • How applications send requests and receive AI-generated responses
Meta Tags	API, AI, Artificial Intelligence, Spoken Tutorial, Python, Machine Learning, Development, Integration, Authentication, Prompts, Requests, Responses
Pre-requisite Tutorial	Basics of Python

Script

Visual Cue	Narration
Title Slide	Welcome to this Spoken Tutorial on Understanding APIs.
Learning Objectives Slide	In this tutorial, you will learn, • Define what an API is in the context of AI. • Explain the role of API keys in secure API access. • Describe how applications send requests and receive AI-generated responses. • Apply the concept of prompts as inputs for AI models.
System Requirements Slide	Here I am using a browser on a computer or mobile.
Pre-requisite Slide	To follow this tutorial, you should have basics of Python .

Illustration showing API with simple icons, connecting two applications	What if I told you different software applications can talk to each other? Imagine you are at a restaurant. You tell the waiter your order. The waiter takes your order to the kitchen. The kitchen prepares your food. Then, the waiter brings your food back to you. Here, the waiter is like an API. An API, or Application Programming Interface, is a software intermediary. It allows two applications to communicate. It takes your request and delivers the response. With that understanding, let's see how APIs work with AI models.
Person at laptop, typing a prompt like 'Write a short story about a talking cat.'	How do you tell an AI model what you want it to do? You send it a prompt. A prompt is your instruction or question. It is the input you give to the AI model. You send these prompts through an API. For example, you might prompt an AI to 'Write a short story about a talking cat.' The API delivers this prompt to the AI model. But how do we ensure only authorized users send these prompts?
Illustration showing a digital key or a key card unlocking a service	To keep APIs secure, we use API keys. Think of an API key as your digital credential. It is a unique code. You send this key with your request. API keys authenticate your requests. They control your access to API services. This ensures only authorized users can interact with the AI. Just like a hotel key card lets you enter your room. Your API key grants you access to the API. Now, let's look at how applications actually send these requests.
Screen close-up: user typing 'tell me something interesting...'	Let's see a simple, but not ideal, way to send a request. Here is a basic Python code snippet. Notice the prompt in this example is very general. It just asks the AI to 'tell me something interesting'. What kind of response do you expect from such a vague prompt? It might be too broad or unhelpful. This request lacks clarity for the AI. Now, let's improve this request for better AI responses.

Screen close-up: user typing 'Explain quantum physics in simple terms.'	Here's an improved example of sending a request. We will use a more specific prompt. See how the prompt is now 'Explain quantum physics in simple terms.' This provides clear instructions to the AI model. We are also specifying the desired length of the response. A clear prompt helps the AI give a relevant answer. It guides the AI to focus on your exact need. Let's compare these two methods side-by-side.
Side-by-side: vague prompt output vs improved prompt output	Look at the vague request on the left. Now, look at the improved request on the right. The vague request gives the AI little direction. The improved request gives clear, specific instructions. It also sets expectations for the response. Structured API interactions lead to better AI responses. Clear prompts save time and improve accuracy. This is crucial when building AI applications. Let us summarize what we learned today.
Summary Slide	Let us summarize what we learned. We defined APIs as intermediaries for applications. We saw how prompts are inputs for AI models. API keys secure access to API services. Finally, we learned to send effective requests to AI.
Assignment Slide	Now as an assignment, Create a simple Python script to send a basic text prompt to an AI model using a hypothetical API. Ensure your prompt is clear and specific. Compare the results of a vague versus a specific prompt.
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